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IARCO RESEARCH PROPOSAL

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Research Title: Quantifying the Macroeconomic Impact of New Generation Space Debris Models on LEO Mega-Constellations

Research Problem

Space debris present a growing threat as their population continues to rise since the 1970s in the Low Earth Orbit. As the number of satellites in Earth's orbit increases, the probability of collisions increases, which leads to the creation of fragments that cause more collisions. This leads to an exponential growth of orbital debris, a phenomenon called the Kessler syndrome [1], which may render Earth's orbit unusable for satellites in the future. This makes the study of long-term consumption of the space environment capacity paramount, so as to assess a risk metric for current and future space missions, predict economic losses and review mitigation actions [9].

To predict the evolution of this debris population in the LEO in order to assess its long-term sustainability, breakup models are developed, NASA's Standard Breakup Model (SBM) being a classical example. It uses a linear trend in logarithmic space to calculate characteristic length of debris, using an EMR (energy-to-mass ratio, impactor kinetic energy divided by system mass) threshold of 40 J/g to split collisions into catastrophic and sub-catastrophic regimes [2]. However, modern-day satellites possess complex geometry and orbital collisions are rarely head-on. The SBM fails to account for both glancing impacts and structural complexity, often leading to overpredictions [3]. To address these limitations, a novel Italian Breakup Model has been proposed by Italian researchers. This model introduces an intermediate regime between 40 J/g and 140 J/g EMRs, and a parameter k_1 to distinguish between central ($k_1 = 1$) and glancing ($k_1 < 1$) impacts [4].

The significance of predicting space debris evolution using physical models is to measure the risk they present in dollars, as it enables direct assessment of mitigation methods such as debris remediation, reduction of the standard 25 year PMD (post-mission disposal) timeline and just in-time collision avoidance [5]. In 2020, expected annual losses from collisions came out to be $86-103 million, over 70% of which are concentrated in the 600-900 km LEO band [6]. Moreover, Kessler syndrome remains a possibility between the years 2040 and 2148 [8], and without any proper mitigation, the cost of orbital debris may soar to more than 0.5% of the world's GDP [7].

While high-fidelity simulation tools exist, century-scale models such as the DISE (Dynamic Integrated Space-Economy) model rely on empirical formulations for space debris dynamics, most notably the legacy NASA SBM [7], which overpredicts debris generation from non-central impacts and most likely lead to inflated values of risk metrics. Therefore, the purpose of this research is to study how the substitution of the NASA SBM with new-generation, high fidelity models, like the Italian Breakup Model, alters these projected losses for LEO mega-constellations, and possibly infer what it means for existing mitigation guidelines.

Existing Literature

The baseline was established by Johnson et. al. with the NASA Standard Breakup Model, which highlights empirical formulations in logarithmic space to calculate characteristic length of space debris, and features a 40 J/g EMR threshold to split collisions into catastrophic and sub-catastrophic regimes [2]. This threshold was analysed by Schimmerohn et. al. using hydrocode simulations of a complex spacecraft colliding with a smaller satellite [3]. This revealed that satellite geometry and collision vector (that determines if the impact is direct or glancing) strongly influence fragmentation, which is not reflected in the empirical formulation of the SBM [3, 4]. As such, Olivieri et. al. proposed a three-section piecewise model that addresses these limitations [4]. The model features a cross-section ratio α to scale fragmentation based on the sizes of the involved bodies, and a parameter k_1 to distinguish between central ($k_1 = 1$) and glancing ($k_1 < 1$) impacts. Model calibration studies confirm the existence of two EMR thresholds: 40 J/g and 140 J/g. Between these thresholds is the intermediate regime between sub-catastrophic and catastrophic collisions. However, this model is still under development as future research is required for area-to-mass ratio and velocity distribution models.

Modern space economics treats space debris as a part of the Tragedy of Commons. Adilov et al. calculated that satellite collisions already cause $86 to 103 million in annual losses, mostly in the crowded 600 to 900 km orbit [6]. From a larger perspective, Bongers et al. used their DISE model to show that without any action, the global cost of space debris could eventually exceed 0.5% of the world's GDP [7]. On the other hand, Rao and Rondina warned that this uncontrolled growth could lead to a disastrous Kessler Syndrome as early as 2040 [8]. To manage this, Letizia et al. created risk metrics to measure how much space capacity is consumed by different missions [9]. However, a major problem exists in these economic studies. They all depend on outdated physical models, like the NASA SBM, to calculate debris risks. Currently, there is no research that combines new generation engineering models, such as the Italian Breakup Model, with economic frameworks. This gap means we cannot accurately measure how better physical models could change the financial risks and economic limits of space operations.

Research Question

"By how much do the macroeconomic liabilities - specifically the Collision Avoidance Maneuver (CAM) expenditures and orbital asset insurance risk - predicted by the NASA Standard Breakup Model vary when substituted by the novel Italian Breakup Model?"

To answer this main question, this research will look into the following sub-questions:

1. How do the physical outputs, such as the count and distribution of fragments, differ between the NASA SBM and the Italian Breakup Model with different impact geometries (central vs glancing impacts)?

2. How do these physical differences lead to measurable differences in projected operational costs, mainly the fuel costs and revenue losses related to Collision Avoidance Maneuvers (CAMs)?
3. How does switching from NASA SBM's rigid 40 J/g catastrophic threshold to the continuous Energy-to-Mass Ratio (EMR) transition proposed by the Italian model change the estimated economic impact of space collisions?

Methodology

The methodology is divided into three phases: (1) defining a controlled, hypothetical scenario of a collision event in the LEO mega-constellation, (2) executing a computational physics pipeline to generate debris size distribution, and (3) translating these physical outputs to quantifiable macroeconomic liabilities.

Phase 1: Scenario Definition

- The research begins with the simulation of a hypervelocity collision in the LEO, 800 km from Earth's surface.
- **Target**: A mega-constellation satellite of mass M = 500 kg and characteristic length $L_{c,T}$ = 2.5 m.
- **Impactor**: A debris of mass m = 1 kg and characteristic length $L_{c,imp}$ = 0.06 m (6 cm). This length is chosen as it sits in the range of 1 - 10 cm, which is the characteristic length of untrackable debris that can cause fatal/catastrophic damage to spacecraft upon impact.
- From the circular orbit baseline assumption ($v_c = \sqrt{(\mu_E/(R_E + h))} = 7.456$ km/s), an impactor at this altitude can reach speeds in the range of 7 to 8 km/s. We consider an impact velocity of 8 km/s for this scenario.
- EMR = Kinetic energy of impactor / Mass of system ≈ 64 J/g. This goes well into the catastrophic regime as per NASA SBM, but sits in the intermediate regime of the Italian model. This calibration ensures that the divergence between these models gets captured by the simulation.

Phase 2: Computational Physics Pipeline

- A computational pipeline is developed using Python to analyse fragmentation using both models.
- **Model A (Legacy baseline)**: This pipeline applies the NASA SBM equations. Since the collision is in the catastrophic regime (EMR ≈ 64 J/g), the model will utilize total system mass (M = 501 kg) to obtain the number distribution of fragments using the standard linear logarithmic trend.
- **Model B (High fidelity model)**: This pipeline applies the Italian Breakup Model equations. The code will calculate transition coefficients (c_1, c_2) and cross-section ratio (α). Furthermore, this code will be executed twice:

  - Case 1: Central impact, $k_1 = 1$ (impactor hits the satellite core)
  - Case 2: Glancing impact, $k_1 = 0.25$ (impactor clips off the edge/solar panel)
- The primary physical output metrics are $\mathbf{N_{>10cm}}$ (space debris of characteristic length more than 10 cm) and $\mathbf{N_{1\text{-}10cm}}$ (space debris of characteristic length 1-10 cm).
- Expected outcome: The NASA SBM pipeline will overpredict these values, especially for the case of a glancing impact.

Phase 3: Economic Cost framework

To convert physical debris results into measurable costs, this study suggests a first-order analytical cost model. While comprehensive Integrated Assessment Models exist for long-term forecasts [7], this study focuses on specific financial risks caused by a particular debris event.

Metric 1: Collision Avoidance Maneuver (CAM) Costs

Debris bigger than 10 cm is tracked by ground radar and triggers conjunction checks, forcing satellites to burn fuel to dodge. The expected cost is calculated as:

$$Cost_{CAM} = \Sigma(N_{>10cm} \times P_{conj} \times C_{maneuver} \times \tau(A/M))$$

Where:

- $N_{>10cm}$ is the count of trackable debris pieces found by the physical model.
- P_{conj} is the probability of conjunction per fragment, based on risk metrics by Letizia [9].
- $C_{maneuver}$ is the baseline financial cost of one avoidance maneuver, involving fuel cost and lost operation time [5].
- $\tau(A/M)$ is defined as an orbital lifetime multiplier. As shown in long-term simulations [9], debris with low (A/M) ratios experience less drag and remain in the orbital shell for decades, which add to the fuel cost over the mega-constellation's operational lifespan.

Metric 2: Expected Insurance Risk

Fragments between 1 cm and 10 cm are untrackable, which gives them mission-ending potential if they collide with a satellite, and this directly drives space insurance premiums. The expected asset loss is modelled as:

$$Risk_{Loss} = N_{1\text{-}10cm} \times P_{lethal} \times V_{asset} \times \kappa(\Delta v)$$

Where:

- $N_{1\text{-}10cm}$ is the count of untrackable debris given by the physical pipeline.
- P_{lethal} is the probability of a fatal collision per fragment.
- V_{asset} is the replacement value of the mega-constellation satellite.

- κ(Δv) is a severity multiplier. It accounts for the velocity distribution of the debris cloud, which affects the relative kinetic energy of future crossing orbits [6].

Scope and Limitations

The area-to-mass (A/M) ratio and velocity (Δv) distributions for the Italian Breakup Model are still under research. To account for this, and to maintain focus on the size and geometry upgrades of the Italian model, the methodology uses the A/M and Δv distribution formulations of the NASA SBM as a proxy for both pipelines.

Research Topic

The advent of mega-constellations in recent years has littered the LEO with debris and is threatening its sustainability. Legacy mitigation guidelines such as the 25-year post mission disposal rule did not account for these constellations, and now they prove to be insufficient to prevent exponential debris growth [9]. With Kessler syndrome being a possibility as early as 2040 [8], the window for accurately pricing and mitigating this risk is closing rapidly, making this research urgently needed.

While the space economics community has made massive developments in quantifying debris risk, most macroeconomic models rely on legacy physical breakup models, and currently no literature exists that integrate a high-fidelity model into a macroeconomic framework, more specifically one that accounts for collision geometry and defines a more continuous EMR threshold [4]. This research topic addresses this very gap and aims to answer how much capital does the space industry save by switching to a new-generation physical breakup engine from legacy models. Moreover, it bridges the physical and economical aspect of space debris and creates a new angle for space sustainability studies.

Quality of Writing

The language of this proposal is academically precise, clear, and appropriate for a multidisciplinary audience. Jargon is minimized, and the structure follows standard research norms. The proposal adheres to grammar, citation, and formatting guidelines strictly to meet IARCO standards.

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